



Written examination in Control of Electrical Drive Systems and Converters

18 June 2025, 09:30 – 13:30 + 15 min. to upload

Allowed aids

All aids are allowed including textbooks, slides, personal notes, exercise solutions, advanced calculators and PC. Use of the internet is allowed in connection with using your aids e.g., OneDrive (off-line mode), course room in Moodle, MATLAB and Simulink software programs.

Use of ChatGPT is strictly forbidden.

General Information

The weightage of each question is mentioned in the parenthesis. The exam set consist of 4 problems.

Questions to the exam questions must always be addressed to the invigilators.

Tablet or mobile phone may be brought to the exam only for use in connection with uploading handwritten pages. During the exam itself, the tablet or phone must be hidden away in a bag, or similar.

When the exam has ended, a maximum of 15 minutes is set aside to upload the exam papers to the Digital Exam System e.g., 13:30-13:45.

Upload your exam papers as a PDF file. Preferably, as 1 common PDF file. Please write your study no. and page number on all pages. Do not write your name as your evaluation is anonymous.

Invigilation is to be contacted in case you want to submit early (but not during the last 30 minutes)

Should you choose to make a handwritten or partly handwritten answer, it must be easy to read. If this is not the case, you may be evaluated as 'not passed'.

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Part 1 Converters (50 %)

Problem 1 (30 %)

- 1) Consider a full-bridge single-phase VSI with unipolar modulation. Draw the schematics, switching states table, waveform for carrier and references, gate signals for S+ and S- and the output voltage V_{an} , V_{bn} and V_{ab} considering sinusoidal PWM. What is the maximum output voltage RMS for the fundamental with respect to input dc voltage. What is voltage THD, its formula and how and why it depends on switching frequency. (15%)
- 2) Space Vector Modulation: Define it and explain why in vector control of ac motor SVM it is preferred against PWM sinusoidal. Draw the space vector representation showing all active vectors in all 6 sectors and the on-time periods T_1 , T_2 and T_0 . Why in the sequence table the order of applying the vectors is mirrored? (15%)

Problem 2 (20 %)

- 1) Thermal Modelling of Switches
 - a. Explain and draw the electrical equivalent circuit of the thermal model for a switch, including the junction, case, and ambient. What is the electric-thermal duality, and why do we use it?
 - b. Consider the MOSFET CSD19532 used in an inverter application. The device experiences a switching loss of 3 W and conducts an RMS current of 6 A. The on-resistance of the MOSFET is 20 m Ω at room temperature.
 - Compute the conduction loss of the MOSFET.
 - Compute the total conduction loss if three identical MOSFETs are connected in parallel.
 - Compute the total power loss (switching + conduction) for the single-MOSFET case.
 - Estimate the junction temperature for the single MOSFET case, assuming no heatsink is used.
- 2) Active Front-End Converters
 - a. Explain the principle and provide a schematic of the active front-end (AFE) converter. Why is it bidirectional, and how can we control the input grid current to be sinusoidal?
 - b. Is the AFE topology used in EV chargers, and what potential future applications could this enable?

Part 2 Electrical Drives (50 %)

Problem 3 (30%)

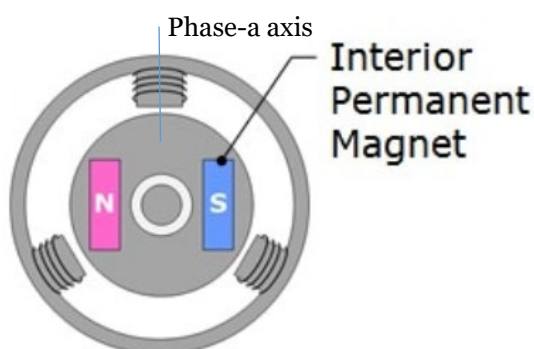
A permanent magnet synchronous machine is modelled in a synchronously rotating dq-reference frame as:

$$\begin{aligned} u_q &= R i_q + p \lambda_q + \omega_r \lambda_d & \lambda_q &= (L_{ls} + L_{mq}) i_q = L_q i_q \\ u_d &= R i_d + p \lambda_d - \omega_r \lambda_q & \lambda_d &= (L_{ls} + L_{md}) i_d + \lambda_{mpm} = L_d i_d + \lambda_{mpm} \end{aligned}$$

The torque equation is

$$T_e = \frac{3}{2} p (\lambda_d i_q - \lambda_q i_d)$$

- In field-oriented control, we introduce back-EMF decoupling into the d-axis current controller. Please provide the control block diagram (including the machine model) to illustrate how this should be done.
- The reference speed is hold constant, and the machine operates in steady state with no load. Suddenly, a load torque is applied. Please discuss the current responses with and without the back-EMF decoupling term during this transient.
- Will the switching frequency affect the bandwidth of the current controller? Is high switching frequency always benefit for achieving a high bandwidth in current controller?
- The rotor position is estimated based on back-EMF. In steady state, the rotor position estimation error is nearly zero. During a transient that a step load is suddenly applied, will the estimated rotor position lead or lag the real rotor position? Please explain Why.
- The following illustrates an interior PMSM with phase resistance $R=2$ (Ohms), inductance $L_d = 2$ (mH), and $L_q = 5$ (mH). It is defined that the center of the rotor north pole aligned with the phase-a axis corresponds to a rotor position of zero. For the initial rotor position alignment, $v_\alpha = 10$ Volts and $v_\beta = 0$ Volts are applied. Please estimate the rotor position (negative, zero, or positive) and explain your reasoning.



Problem 4 Induction machine (20%)

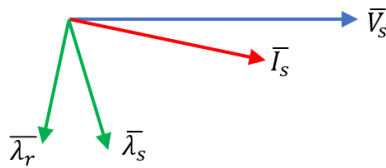
The mathematical model of an induction machine may be expressed as:

The stator side:

The rotor side:

$$\begin{aligned}
 u_{qs} &= r_s i_{qs} + p \lambda_{qs} + \omega_e \lambda_{ds} & \lambda_{qs} &= L_{ls} i_{qs} + L_m (i_{qs} + i_{qr}) & u_{qr} &= r_r i_{qr} + p \lambda_{qr} + (\omega_e - \omega_r) \lambda_{dr} & \lambda_{qr} &= L_{lr} i_{qr} + L_m (i_{qs} + i_{qr}) \\
 u_{ds} &= r_s i_{ds} + p \lambda_{ds} - \omega_e \lambda_{qs} & \lambda_{ds} &= L_{ls} i_{ds} + L_m (i_{ds} + i_{dr}) & u_{dr} &= r_r i_{dr} + p \lambda_{dr} - (\omega_e - \omega_r) \lambda_{qr} & \lambda_{dr} &= L_{lr} i_{dr} + L_m (i_{ds} + i_{dr})
 \end{aligned}$$

where λ represents the flux linkage. The voltage supplied to the rotor windings is zero, assuming it is a squirrel cage induction machine.



- Now we will implement indirect rotor flux-oriented control. Please draw the dq reference frame where this controller will be based on.
- Please simplify the above equations to derive the machine equations for realizing the indirect rotor flux-oriented control. You do not need to give all the intermediate steps. But it is important to mention:
 - to use which equation(s) (e.g. stator/rotor voltage/flux linkage equations);
 - using the selected equations, to get rid of or obtain which variable as a function of which variables;
 - the final result.
 - controller topology derived based on the obtained equations.



Study number:	Programme: EN8
Ordinary examination: Control of Electrical Drive Systems and Converters Tuesday 18 June 2024 at 9:30-13:30 + 15 minutes (13:45)	

- Please write your study no. and page number on all pages. Do not write your name as your evaluation is anonymous!
- **All kinds of communication with other students or the outside world is strictly forbidden. Use of ChatGPT is strictly forbidden.**
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- Upload your answer sheets as a PDF file, preferably as 1 common file.
- Your answers must be uploaded to Digital during the last quarter from 13:30-13:45. After 13:30 you are no longer allowed to answer any of the assignments. Invigilation is to be contacted in case you want to submit early (but not during the last 30 minutes).
- It is not allowed to access recorded lectures during the examination.
- This examination set contains **4** problems.
- You are welcome to take the exam papers/questions with you after the exam has finished. However, you are NOT allowed to take them with you if you leave before the exam has ended.



Part 1 Converters (50 %)

Problem 1 (30 %)

- 1) Consider a half-bridge single-phase VSI. Draw the schematics, switching states table, waveform for carrier and references, gate signals for S+ and S- and the output voltage considering sinusoidal PWM. What is the maximum output voltage RMS with respect to input voltage. (15%)
- 2) Space Vector Modulation: Define it and explain why it is more suitable for vector control of ac machines in comparison to PWM sinusoidal. Draw the space vector representation showing all active vectors in different sectors with the formulas of state times for the two active vectors and zero vector for sector 1. Draw the switching sequence table for sector 2 and the resulting generic gate signals for the three phases. (15%)

Problem 2 (20 %)

- 1) Answer the following:
A MOSFET has the following parameters:
 $V_{plateau} = 8V, V_{th} = 4V, R_G = 5\Omega, V_{DC,blocking} = 70V, I_D = 25A, C_{ISS} = 6nF, C_{RSS} = 350pF, f_{sw} = 10kHz, I_{RMS} = 7A$, on – resistance ($r_{d(on)}$) = 5 mΩ
 - (a) Assume the turn-on loss is similar to the turn-off loss. Calculate the total switching loss (5%)
 - (b) Calculate the conduction loss. Using the switching loss in part (a) above, compute the total losses. What should be the total thermal resistance to the ambient (from junction to ambient) if the allowable temperature rise is 50°C (5%)
- 2)
 - a) Explain the working principle of a Direct Matrix Converter (DMC) (5%)
 - b) Explain why the switching frequency changes in a boost Power Factor Correction (PFC) converter operating in Critical Conduction Mode (CrCM) (5%)

Part 2 Electrical Drives (50 %)

Problem 3 (30%)

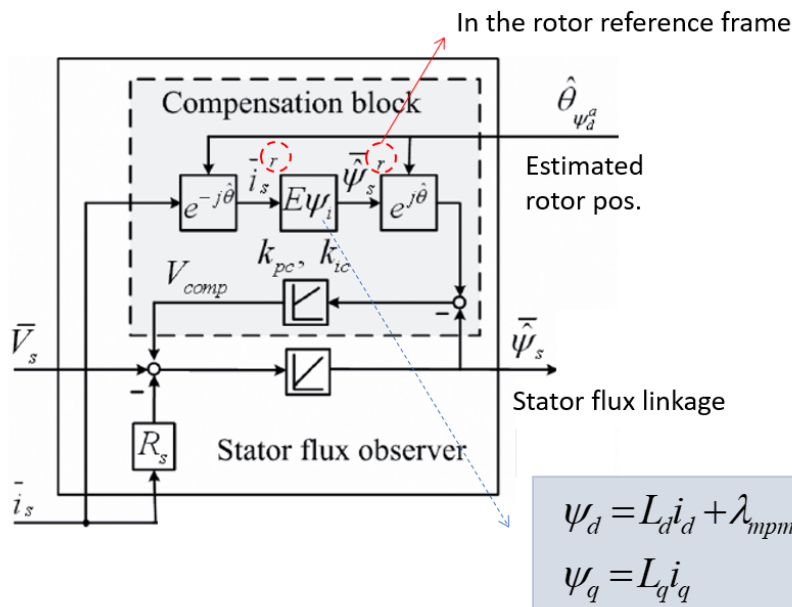
A permanent magnet synchronous machine is modelled in a synchronously rotating dq-reference frame as:

$$\begin{aligned} u_q &= R i_q + p \lambda_q + \omega_r \lambda_d & \lambda_q &= (L_{ls} + L_{mq}) i_q = L_q i_q \\ u_d &= R i_d + p \lambda_d - \omega_r \lambda_q & \lambda_d &= (L_{ls} + L_{md}) i_d + \lambda_{mpm} = L_d i_d + \lambda_{mpm} \end{aligned}$$

where $L_d = L_q = L_s$. The torque equation is

$$T_e = \frac{3}{2} p (\lambda_d i_q - \lambda_q i_d)$$

- We introduce back-EMF decoupling to the q-axis current controller. Please provide the control block diagram (including the machine model) to illustrate how this should be done.
- Please discuss what the differences (e.g. steady state error, dynamic tracking error, etc.) between the current controllers with and without the back-EMF decoupling term.
- The following closed-loop flux observer was implemented in the laboratory.

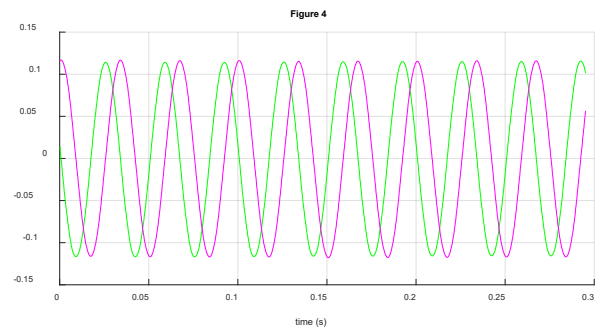
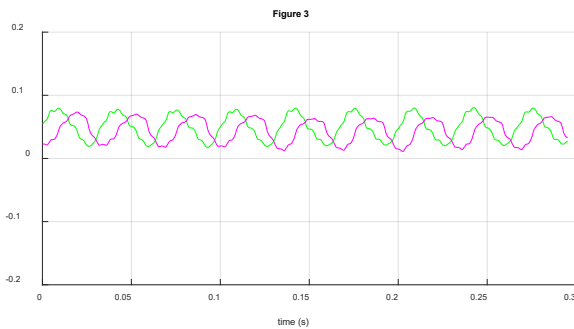
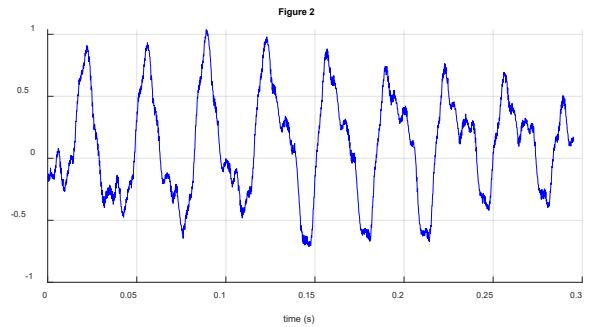
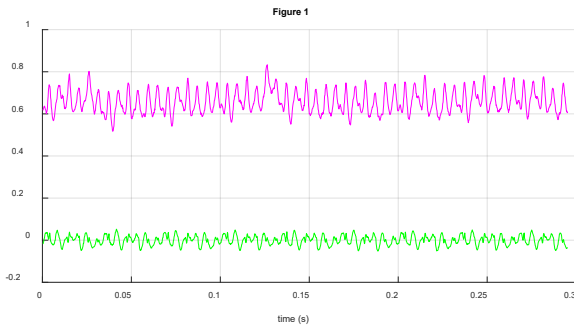


The following four figures contain the experimentally recorded waveforms. It is known that figure 2 below shows the position estimation error in electrical degrees. The remaining three figures show:

- $\alpha\beta$ – components of the estimated stator flux linkages.
- $\alpha\beta$ – components of the “ V_{comp} ” signal indicated in the figure above.
- dq – components of the machine current.

Please explain what signals figures 1, 3 and 4 are showing, respectively, and explain why?

- d) Continued from c), knowing that the machine has 4 pole pairs, what is the mechanical speed in rpm for this operation condition?
- e) Continued from c), now assume the q-axis current is zero (ideal no-load condition). Please calculate the values for λ_d, λ_q ?



- f) Instead of Field-Oriented Control (FOC), sometimes we use direct torque control (DTC). For DTC, we typically use the following torque equation. It is claimed that by controlling the machine terminal voltage vector, we can immediately increase or decrease the torque. Please explain briefly why this is achievable?

$$T_e \propto \left| \bar{\lambda}_s \times \bar{\lambda}_r \right| = \left| \bar{\lambda}_s \right| \left| \bar{\lambda}_r \right| \sin(\theta_{sr})$$

where θ_{sr} is the angle between the stator flux linkage vector and the rotor flux linkage vector.

Problem 4 Induction machine (20%)

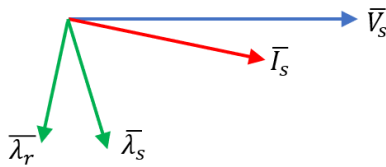
The mathematical model of an induction machine may be expressed as:

The stator side:

The rotor side:

$$\begin{aligned}
 u_{qs} &= r_s i_{qs} + p \lambda_{qs} + \omega_e \lambda_{ds} & \lambda_{qs} &= L_{ls} i_{qs} + L_m (i_{qs} + i_{qr}) & u_{qr} &= r_r i_{qr} + p \lambda_{qr} + (\omega_e - \omega_r) \lambda_{dr} & \lambda_{qr} &= L_{lr} i_{qr} + L_m (i_{qs} + i_{qr}) \\
 u_{ds} &= r_s i_{ds} + p \lambda_{ds} - \omega_e \lambda_{qs} & \lambda_{ds} &= L_{ls} i_{ds} + L_m (i_{ds} + i_{dr}) & u_{dr} &= r_r i_{dr} + p \lambda_{dr} - (\omega_e - \omega_r) \lambda_{qr} & \lambda_{dr} &= L_{lr} i_{dr} + L_m (i_{ds} + i_{dr})
 \end{aligned}$$

where λ represents the flux linkage. The voltage supplied to the rotor windings is zero, assuming it is a squirrel cage induction machine.



- Implement stator flux-oriented control. Please add the dq-frame for stator flux-oriented control to the above figure.
- What is the torque equation to be based on for implementing the stator flux-oriented control?
- Please derive the equation for calculating the slip speed under stator flux-oriented control scheme.
- In generalized flux-oriented control, the general flux vector is defined as

$$\bar{\lambda}_{ma} = a L_m \left(\bar{i}_s + \frac{\bar{i}_r}{a} \right).$$

Please prove that when $a = \frac{L_m}{L_r}$, the above general flux vector will be aligned with the rotor flux vector.

Examination date: 16-06-2023

Elaborated by: Remus Teodorescu, Abhijit Kulkarni (Part1); Kaiyuan Lu (Part 2)

Reviewed by : Kaiyuan Lu (Part1); Remus Teodorescu, Abhijit Kulkarni (Part 2)

Written examination in Control of Electrical Drive Systems and Converters

8th semester PED 2023

-
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 - This exam contains **2 main** parts (4 questions in total). The percentage is indicated for every question.
-

Part 1 Converters (50 %)

Problem 1 (30 %)

- 1) Consider a full-bridge single-phase VSI. Draw the schematics, switching states table, waveform for carrier and references, gate signals for S1+ and S2+ and the output voltage considering unipolar sinusoidal PWM. How is the switching harmonics in the output voltage with respect to the carrier frequency? (15%)
- 2) Space Vector Modulation: Define it and explain why it is more suitable for vector control of ac machines in comparison to PWM sinusoidal. Draw the space vector representation showing all active vectors also on sector with the formulas of state times for the two active vectors and zero vector. Draw the switching sequence table for sector 1 and the resulting generic gate signals for this sector. (15%)

Problem 2 (20 %)

- 1) Answer the following:
 - (a) An IGBT has $E_{on} = 4mJ$ and $E_{off} = 2mJ$. These energies are defined in the datasheet at a test voltage of $V_t = 600V$ and a test current of $I_t = 20A$.
It is used in a power converter with the IGBT blocking voltage of $V_{dc} = 400V$ and the IGBT average current $I_{avg} = 20A$. If the switching frequency is 3kHz, what is the total switching loss? (5%)
 - (b) Consider that the conduction loss for the same IGBT is 8W. Using the switching loss in part (a) above, compute the total losses. What should be the total thermal resistance to the ambient (from junction to ambient) if the temperature rise from ambient for the IGBT under consideration should be less than $60^\circ C$? (5%)
- 2)
 - (a) Explain how a boost power factor correction (PFC) circuit is controlled in critical conduction mode (CrCM). Draw the circuit diagram of the boost PFC along with the control circuit used for implementing the closed-loop control in CrCM mode. (5%)
 - (b) What is the expression for the duty ration in a boost PFC with continuous conduction mode? (5%)

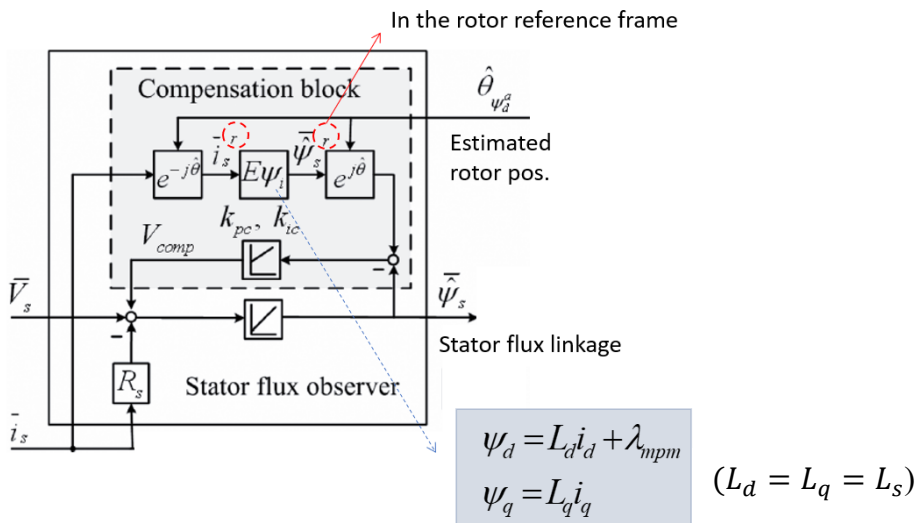
Part 2 Control of electrical machines (50 %)

Problem 3 Permanent magnet machine (25%)

The machine equations for a surface mounted permanent magnet machine are given below.

$$\begin{aligned} u_q &= R i_q + p \lambda_q + \omega_r \lambda_d & \lambda_q &= (L_{ls} + L_{mq}) i_q = L_q i_q \\ u_d &= R i_d + p \lambda_d - \omega_r \lambda_q & \lambda_d &= (L_{ls} + L_{md}) i_d + \lambda_{mpm} = L_d i_d + \lambda_{mpm} \end{aligned}$$

- It is desired to park the rotor at 30 electrical degrees. You will control the DC currents in phase a-, b- and c-windings to achieve this. It is found that phase-a current is 10 A, please determine the phase-b and phase-c currents respectively and explain why. Please also determine the voltage commands $v_\alpha = ?$ $v_\beta = ?$
- For a field-oriented controller, without the back-EMF decoupling term, for the q-axis current controller, please give the transfer function of
$$\frac{\text{shaft mechanical speed}}{\text{load torque}} = \frac{\omega_{r.mec}}{T_L} = ?$$
 (Assuming the reference q-axis current is zero.)
- For back-EMF based sensorless control, the following flux observer is implemented. Please discuss:
 - What is the purpose of the “compensation block” indicated in the following figure.
 - How does this “compensation block” work?
 - If the inductance used in this “compensation block” is higher than the actual motor inductance, what is the consequence?



Problem 4 Induction machine (25%)

The mathematical model of an induction machine may be expressed as:

The stator side:

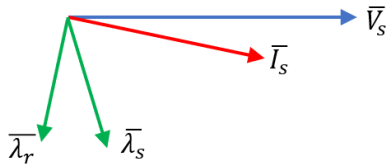
$$\begin{aligned} u_{qs} &= r_s i_{qs} + p \lambda_{qs} + \omega_e \lambda_{ds} & \lambda_{qs} &= L_{ls} i_{qs} + L_m (i_{qs} + i_{qr}) \\ u_{ds} &= r_s i_{ds} + p \lambda_{ds} - \omega_e \lambda_{qs} & \lambda_{ds} &= L_{ls} i_{ds} + L_m (i_{ds} + i_{dr}) \end{aligned}$$

The rotor side:

$$\begin{aligned} u_{qr} &= r_r i_{qr} + p \lambda_{qr} + (\omega_e - \omega_r) \lambda_{dr} & \lambda_{qr} &= L_{lr} i_{qr} + L_m (i_{qs} + i_{qr}) \\ u_{dr} &= r_r i_{dr} + p \lambda_{dr} - (\omega_e - \omega_r) \lambda_{qr} & \lambda_{dr} &= L_{lr} i_{dr} + L_m (i_{ds} + i_{dr}) \end{aligned}$$

where λ represents the flux linkage. The voltage supplied to the rotor windings is zero, assuming it is a squirrel cage induction machine.

In steady state, the vectors of the stator voltage, stator current, stator and rotor flux linkages may be drawn as below. You will implement an indirect rotor flux-oriented control (meaning there is no shaft encoder and the rotor speed and position cannot be directly measured).



- Please add the dq-frame for rotor flux-oriented control to the above figure. Please a) derive the expression of the rotor flux linkage as a function of stator currents; b) simplify this expression for steady state operation.
- The rotor flux vector will be estimated from the stator flux, which can be calculated from the terminal voltage command. Please show how the rotor flux may be found from the stator flux linkage.
- Please give the controller block diagram for the indirect rotor-flux oriented control for realizing closed-loop speed control, i.e., torque and flux needs to be controlled separately. Suppose the rated rotor flux is known. Please derive the equations first (i.e., how the rotor flux may be controlled by stator d-axis current, etc.) and draw the controller block diagram correspondingly.



Study number:	Programme: EN8
Ordinary examination: Control of Electrical Drive Systems and Converters Thursday 9 June 2022 at 9:30-13:30 + 15 minutes (13:45)	

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Aids that are allowed: *textbooks, slides, personal notes, exercise solutions, calculators, and PC.*

Books bought online may be used during the exam as download.
Personal notes slides saved under OneDrive are allowed. Synchronisation must be turned off/paused during the exam. This must be visible from the taskbar on your PC. Notes and/or slides saved as a pdf document must be opened via Adobe – Not in an online browser.

It is NOT allowed to be online during the exam. However, it is allowed to access:

- Digital Exam System
- Moodle course room

Tablet or mobile may be brought to the exam for use in connection with uploading handwritten pages. When the exam has ended, a maximum of 15 minutes is set aside to upload the exam papers to the Digital Exam System e.g., 13:30-13:45.

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Written exam in Control of Electrical Drive Systems and Converters

9th June 2022, 09:30-13:30 + 15 minutes

8th semester PED 2022

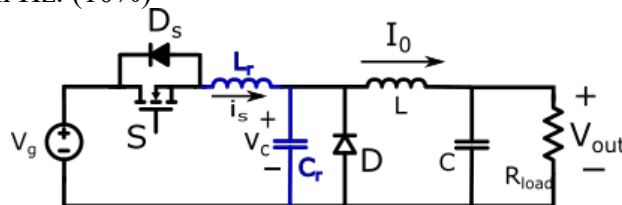
-
- The complete description of the solution of the questions must be attached separately.
 - Only results with a complete description of the solution will be taken into account.
 - All intermediate steps and calculations should be included in your answer sheets --- printing the final result is **NOT** sufficient.
 - This exam contains **2 main** parts (4 questions in total). The percentage is indicated for every question.
-

Elaborated by: Remus Teodorescu (Part1); Kaiyuan Lu (Part 2 with three problems)

Reviewed by : Kaiyuan Lu (Part1); Remus Teodorescu (Part 2)

Part 1 Converters (50 %)

- 1.1 Explain in your own words what SVM is and why it is suitable for vector control of ac machines in comparison to PWM sinusoidal. Use a space vector diagram to sustain your answer. What is the switching sequence in SVM and draw a generic pulse sequence for Sector 1 (10%)
- 1.2 Explain in your words, the difference between using linear current controllers and hysteresis control in terms of: BW, harmonic spectrum, ease of implementation. How can the switching frequency be made quasi-constant with hysteresis control? (10%)
- 1.3 Answer the following:
 - (a) An IGBT has $E_{on} = 3mJ$ and $E_{off} = 2mJ$. It is used in a power converter with the same test voltage and currents at which these energies were specified in the datasheet of the IGBT.
If the switching frequency is 4kHz, what is the total switching loss? (5%)
 - (b) By modifying the topology, it is ensured that the IGBT turns ON at zero voltage, but it still has the same amount of turn-off losses as for the specified test voltage. For this case, what should be the switching frequency to achieve the same amount of total switching losses as in case (a) above? (5%)
 - (c) We would like to mount a heat sink to the IGBT. Is it better to have a smaller thermal resistance for the heatsink or larger thermal resistance? Explain why by using the thermal equivalent circuit. (5%)
- 1.4 How is the output voltage controlled in a ZCS buck converter? What makes it different compared to the hard switched buck converter? (5%)
- 1.5 Explain the different operating modes for ZCS buck converter along with the waveforms of resonant inductor (L_r) current and resonant capacitor (C_r) voltage. Write the expression for the resonance frequency in Hz. (10%)



Part 2 Control of electrical machines (50 %)

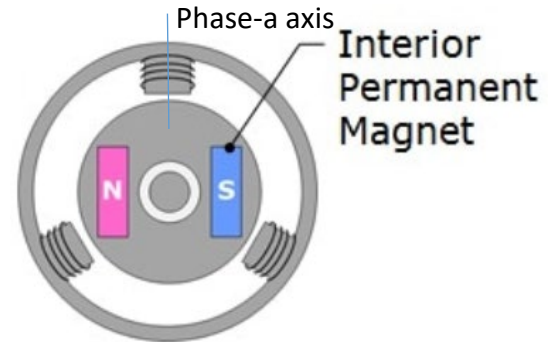
2.1 Permanent magnet machine (30%)

An interior permanent magnet machine is sketched below, modelled in a synchronously rotating dq-reference frame as:

$$\begin{aligned} u_q &= R i_q + p \lambda_q + \omega_r \lambda_d & \lambda_q &= (L_{ls} + L_{mq}) i_q = L_q i_q \\ u_d &= R i_d + p \lambda_d - \omega_r \lambda_q & \lambda_d &= (L_{ls} + L_{md}) i_d + \lambda_{mpm} = L_d i_d + \lambda_{mpm} \end{aligned}$$

The torque equation is

$$T_e = \frac{3}{2} p (\lambda_d i_q - \lambda_q i_d)$$



where p in the torque equation is the number of pole pairs and λ represents the flux linkage.

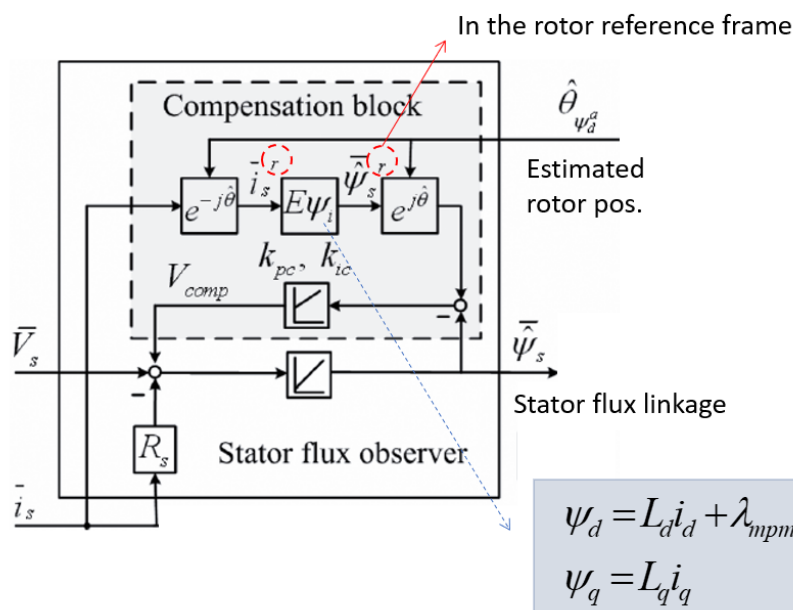
1. Please add the dq-reference frame to the rotor shown above, and explain why. The rotor is assumed to rotate in the **clockwise** direction. The phase-a axis is given. Please indicate the rotor position in the above figure. The phase resistance is $R=0.5$ (Ohms) and the inductance $L_q = 5$ (mH). Should the L_d inductance be smaller or larger than L_q ? Why?
2. Now you are supplying DC voltages $v_\alpha = 15$ Volts and $v_\beta = 10$ Volts to the three phase windings. Suppose the resultant torque is large enough to overcome the friction. Please tell where is the rotor position at steady state (with the supplied DC voltages present). What are the DC phase-a, -b and -c currents?
3. For field oriented control of PM machine, please give the control block diagram in order to control the d-axis current to follow a reference current command; add the back-EMF decoupling term to this control block diagram.
4. Suppose we do not implement the back-EMF decoupling term in your controller. Now we want to examine how the back-EMF (regarded as a disturbance) may affect the controller performance. Please derive the transfer function with the input as the machine back-EMF voltage while the output is the machine d-axis current (assuming the reference i_d current is zero). Please discuss:

- in steady state, what will be the performance difference (meaning the accuracy in tracking the reference) with/without the back-EMF decoupling term in your controller;
- in dynamic situation without this back-EMF decoupling term in your controller, how will you change the PI parameters (increase or decrease) to suppress the disturbance caused by the machine back-EMF term?

2.2 Position estimation (5%)

In lecture 2, slide P8, we discussed the following adaptive mechanism for DC drift compensation in the back-EMF based position observer.

- Please comment where do the DC offsets come from in the integrated flux linkages?
- Please sketch roughly the waveform of signal V_{comp} in steady state.



2.3 Induction machine (15%)

The mathematical model of an induction machine may be expressed as:

The stator side:

$$u_{qs} = r_s i_{qs} + p \lambda_{qs} + \omega_e \lambda_{ds}$$

$$\lambda_{qs} = L_{ls} i_{qs} + L_m (i_{qs} + i_{qr})$$

$$u_{ds} = r_s i_{ds} + p \lambda_{ds} - \omega_e \lambda_{qs}$$

$$\lambda_{ds} = L_{ls} i_{ds} + L_m (i_{ds} + i_{dr})$$

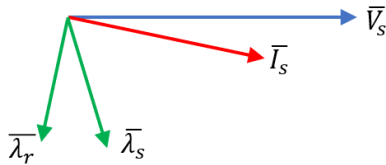
The rotor side:

$$u_{qr} = r_r i_{qr} + p \lambda_{qr} + (\omega_e - \omega_r) \lambda_{dr} \quad \lambda_{qr} = L_{lr} i_{qr} + L_m (i_{qs} + i_{qr})$$

$$u_{dr} = r_r i_{dr} + p \lambda_{dr} - (\omega_e - \omega_r) \lambda_{qr} \quad \lambda_{dr} = L_{lr} i_{dr} + L_m (i_{ds} + i_{dr})$$

where λ represents the flux linkage. The voltage supplied to the rotor windings is zero, assuming it is a squirrel cage induction machine.

In steady state, the vectors of the stator voltage, state current, stator and rotor flux linkages may be drawn as below.



1. Now we will implement the rotor flux-oriented control. Please draw the dq reference frame where this controller will be based on.
2. Please simplify the above equations to derive the machine equations in steady state for realizing the rotor flux-oriented control. You do not need to give all the intermediate steps. But it is important to mention:
 - a), to use which equation(s) (e.g. stator/rotor voltage/flux linkage equations);
 - b), using the selected equations, to get rid of or obtain which variable as a function of which variables;
 - c), the final result.
 - d), controller topology derived based on the obtained equations.

Written examination in Control of Electrical Drive Systems and Converters

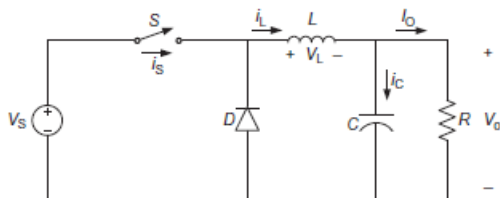
8th semester

-
- All usual helping aids are allowed, including text books, slides, personal notes, and exercise solutions.
 - Calculators and laptop computers are allowed.
 - Communicating with the lecturers in MS-Teams room is allowed.
 - Any kind of communication with others is strictly forbidden.
 - Remember to write your study number on **all** answer sheets.
 - Intermediate steps and derivations/calculations must be included in your answer sheets. Only results with a complete description of the solution will be taken into account.
 - The pages which you upload must be easy to read. Otherwise, answers may be evaluated as “not passed
 - All intermediate steps and calculations should be included in your answer sheets --- printing the final result is **NOT** sufficient.
 - This exam contains **3** parts. The percentage is indicated for every question.
-

Elaborated by: Remus Teodorescu (Part1); Kaiyuan Lu (Part 2 with two problems)
Reviewed by : Kaiyuan Lu (Part1); Remus Teodorescu (Part 2 with two problems)

Part 1 Space Vector Modulation (50 %)

- 1.1 Explain in your own words what SVM is why is suitable for vector control of ac machines. Use a space vector diagram to sustain your answer. What is the switching sequence in SVM. Draw a generic pulse sequence for Sector 1(10%)
- 1.2 What are the advantages/disadvantages of SVM in comparison with SPWM (10%)
- 1.3 What is soft-switching and make a comparison with hard switching in terms of advantages/disadvantages. Explain ZVS using concept diagram. (10%)
- 1.4 Please demonstrate how you can add ZVS to the buck converter shown below (20%):



Draw the waveforms of switch S current/voltage. Derive expressions for peak current/voltage of the switch S and compare with the conventional buck converter.

Part 2 Control of electrical machines (50 %)

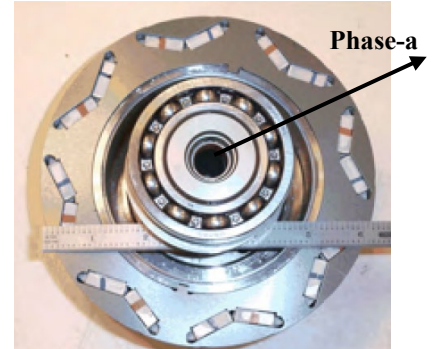
2.1 Permanent magnet machine (30%)

The rotor of an interior permanent magnet synchronous machine used in the hybrid car Prius (2004) is shown below. Its machine model may be represented in a synchronously rotating dq-reference frame as:

$$\begin{aligned} u_q &= R i_q + p \lambda_q + \omega_r \lambda_d & \lambda_q &= (L_{ls} + L_{mq}) i_q = L_q i_q \\ u_d &= R i_d + p \lambda_d - \omega_r \lambda_q & \lambda_d &= (L_{ls} + L_{md}) i_d + \lambda_{mpm} = L_d i_d + \lambda_{mpm} \end{aligned}$$

The torque equation is

$$T_e = \frac{3}{2} p (\lambda_d i_q - \lambda_q i_d)$$



where p in the torque equation is the number of pole pairs and λ represents the flux linkage.

1. What is the number of pole pairs of this machine? The rotor is assumed to rotate in the anti-clockwise direction. Please add the dq-reference frame to the rotor shown above, and explain why. The phase-a axis is given in the figure above. Please indicate the rotor position in the above figure. Should the L_d inductance be smaller or larger than L_q ? Why?
2. Now you want to force the rotor to be parked at 60 electrical degrees, by feeding DC currents to phase-a, -b and -c windings, respectively. Suppose phase-a current is 10 A, then what currents you will supply to phase-b and phase-c respectively?
3. For field-oriented control of PM machine, please give the control block diagram in order to control the q-axis current to follow a reference current command; then add the back-EMF decoupling term to this control block diagram.
4. In the field-oriented control, for this machine, how will you choose a d-axis current reference value (zero, positive or negative)? Why?
5. The machine is in field-oriented control mode and is operating in steady state. Please **sketch** the speed and i_d , i_q current transient waveforms during a sudden step load on event (meaning a step load torque is suddenly added). You should align the responses of the speed, i_q , and i_d using the same time axis, and give brief explanations.

2.2 Induction machine (20%)

The mathematical model of an induction machine may be expressed as:

The stator side:

$$u_{qs} = r_s i_{qs} + p \lambda_{qs} + \omega_e \lambda_{ds}$$

$$\lambda_{qs} = L_{ls} i_{qs} + L_m (i_{qs} + i_{qr})$$

$$u_{ds} = r_s i_{ds} + p \lambda_{ds} - \omega_e \lambda_{qs}$$

$$\lambda_{ds} = L_{ls} i_{ds} + L_m (i_{ds} + i_{dr})$$

The rotor side:

$$u_{qr} = r_r i_{qr} + p \lambda_{qr} + (\omega_e - \omega_r) \lambda_{dr} \quad \lambda_{qr} = L_{lr} i_{qr} + L_m (i_{qs} + i_{qr})$$

$$u_{dr} = r_r i_{dr} + p \lambda_{dr} - (\omega_e - \omega_r) \lambda_{qr} \quad \lambda_{dr} = L_{lr} i_{dr} + L_m (i_{ds} + i_{dr})$$

where λ represents the flux linkage. The voltage supplied to the rotor windings is zero, assuming it is a squirrel cage induction machine.

1. You will implement the rotor-flux oriented FOC for this induction machine. Please give the stator, rotor voltage equations and the torque equation expressed in the rotor flux-oriented reference frame.
2. For the given induction machine listed below, please calculate the rated d-axis current reference value to be used in the rotor flux oriented FOC.

V. Δ/Y	Hz	kW	rpm	A. Δ/Y	COS ϕ
230/400	50	0.3	905	1.9/1.1	0.64

Stator and rotor resistances

$$R_s = 26.5 \text{ (}\Omega\text{)}$$

$$R_r = 23.9 \text{ (}\Omega\text{)}$$

Stator and rotor self-inductances

$$L_s = 0.79 \text{ (H)}$$

$$L_r = 0.79 \text{ (H)}$$

Magnetization inductance

$$L_m = 0.71 \text{ (H)}$$

Written examination in Control of Electrical Drive Systems and Converters

8th semester

-
- All usual helping aids are allowed, including text books, slides, personal notes, and exercise solutions.
 - Calculators and laptop computers are allowed, provided that all wireless and wired communication equipment is turned off.
 - Internet access is strictly forbidden.
 - Any kind of communication with other students is not allowed.
 - Remember to write your study number on **all** answer sheets.
 - The complete description of the solution of the questions must be attached separately.
 - Only results with a complete description of the solution will be taken into account.
 - All intermediate steps and calculations should be included in your answer sheets --- printing the final result is **NOT** sufficient.
 - This exam contains **3** parts. The percentage is indicated for every part.
-

Part 1 (50 %)

- 1.1 How is soft switching realized and what are the advantages and disadvantages compared to hard switching? Use also the switching trajectory in your explanation (10%)
- 1.2 How is three-phase PWM implemented in an analog application? (10%)
- 1.3 Discuss in detail Table 14-2 from the book 'Power Electronics Converters, Application and Design. N. Mohan, T.M. Undeland and W.P. Robbins (cover at least 6 parameters). (10%)

Table 14-2 Comparison of Adjustable Frequency Drives

<i>Parameter</i>	<i>PWM</i>	<i>Square Wave</i>	<i>CSI</i>
Input power factor	+	—	— —
Torque pulsations	++	—	—
Multimotor capability	+	+	—
Regeneration	—	—	++
Short-circuit protection	—	—	++
Open-circuit protection	+	+	—
Ability to handle undersized motor	+	+	—
Ability to handle oversized motor	—	—	—
Efficiency at low speeds	—	+	+
Size and weight	+	+	— —
Ride-through capability	+	—	—

- 1.4 What is discontinuous PWM and what are the advantages and disadvantages of such modulation techniques compared to ST-PWM? How can these be implemented? (20%)

For all questions: besides explaining with your own words, use also figures in your answers.

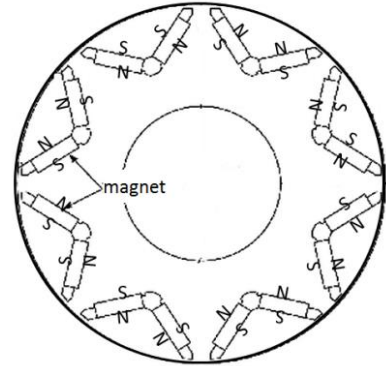
Part 2 (30%)

An interior permanent magnet machine's rotor is shown below. Modelled in a synchronously rotating dq-reference frame, we have:

$$\begin{aligned} u_q &= R i_q + p \lambda_q + \omega_r \lambda_d & \lambda_q &= (L_{ls} + L_{mq}) i_q = L_q i_q \\ u_d &= R i_d + p \lambda_d - \omega_r \lambda_q & \lambda_d &= (L_{ls} + L_{md}) i_d + \lambda_{mpm} = L_d i_d + \lambda_{mpm} \end{aligned}$$

The torque equation is

$$T_e = \frac{3}{2} p (\lambda_d i_q - \lambda_q i_d)$$



where p in the torque equation is the number of pole pairs and λ represents the flux linkage.

1. Please add the dq-reference frame to the rotor shown above, and explain why. The rotor is assumed to rotate in **clockwise** direction (*important to notice this!*).
2. Please determine for this rotor, if ($L_d > L_q$) or ($L_d < L_q$), and explain why. In the dq-frame you have drawn above, please indicate the desired location for the current vector (a rough indication with respect to the q-axis is enough), and explain why.
3. For field oriented control of PM machine, please give the control block diagram in order to control the d-axis current to follow a reference current command; add the back-EMF decoupling term to this control block diagram; simply the control block diagram considering back-EMF decoupling effects.
4. Please indicate the back-EMF term appearing in the q-axis voltage equation; what is the back-EMF decoupling term that should be introduced to the q-axis current regulation block diagram? Please use the machine equations to give mathematical proof that supports your statement.
5. Knowing $R=0.2$ (Ohm), $L_d = 2$ (mH). The inverter switches at 5 kHz and the equivalent delay the inverter introduces to the controller is considered to be 1.5/5000 (s). In the PI controller, $K_p = 3$ and $K_i = 100$. Please calculate the bandwidth of this d-axis current control loop. Please give the bandwidth in Hz.

Part 3 (20%)

The mathematical model of an induction machine may be expressed as:

The stator side:

$$u_{qs} = r_s i_{qs} + p \lambda_{qs} + \omega_e \lambda_{ds}$$

$$\lambda_{qs} = L_{ls} i_{qs} + L_m (i_{qs} + i_{qr})$$

$$u_{ds} = r_s i_{ds} + p \lambda_{ds} - \omega_e \lambda_{qs}$$

$$\lambda_{ds} = L_{ls} i_{ds} + L_m (i_{ds} + i_{dr})$$

The rotor side:

$$u_{qr} = r_r i_{qr} + p \lambda_{qr} + (\omega_e - \omega_r) \lambda_{dr} \quad \lambda_{qr} = L_{lr} i_{qr} + L_m (i_{qs} + i_{qr})$$

$$u_{dr} = r_r i_{dr} + p \lambda_{dr} - (\omega_e - \omega_r) \lambda_{qr} \quad \lambda_{dr} = L_{lr} i_{dr} + L_m (i_{ds} + i_{dr})$$

where λ represents the flux linkage. The voltage supplied to the rotor windings is zero, assuming it is a squirrel cage induction machine.

1. You will implement rotor-flux oriented control for this induction machine. Please give the stator, rotor voltage equations and the torque equation expressed in the rotor flux-oriented reference frame.
2. Please list the motor parameters needed in realization of the rotor-flux oriented FOC.
3. For the given induction machine listed below, please calculate the rated d-axis current reference value to be used in the rotor flux oriented FOC.

V. Δ/Y	Hz	kW	rpm	A. Δ/Y	COS ϕ
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